



LUND
UNIVERSITY

Written Examination
Linear Analysis, MATB24
November 17, 2021
Time: 13.00–18.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

You may freely use formulas from the formula sheet. Refer to relevant theorems from the course when you use them. Write clearly with concise motivations and explain your steps. In each of the six problems a maximum of five points can be achieved. No aids other than writing utensils are allowed.

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(t) = |\sin t|$.
- a) Compute the real Fourier series of f and prove that it converges uniformly to f .
 - b) Determine the value of

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}.$$

2. Find a solution $u = u(x, t)$ for $(x, t) \in [0, \pi/2] \times [0, \infty)$ to the following problem for the heat equation:

$$\begin{cases} 3u_t = u_{xx}, & 0 < x < \pi/2, t > 0, \\ u_x(0, t) = u(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = \cos^3(3x), & 0 < x < \pi/2. \end{cases}$$

Also briefly discuss the regularity of your solution and in which sense the initial condition is satisfied for it.

3. a) Prove that (K_n) , given by

$$K_n: \mathbb{R} \rightarrow \mathbb{R}, \quad K_n(x) = \frac{n}{\pi(1 + n^2x^2)}, \quad n \in \mathbb{N},$$

is a positive summation kernel on $\mathbb{R}$.

- b) Compute

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \frac{ne^{-x^2}}{1 + n^2x^2} dx.$$

4. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$,

$$f(t) = e^{-3|t|}, \quad g(t) = \begin{cases} |t|, & |t| \leq 1, \\ 0, & |t| > 1. \end{cases}$$

- a) Compute the Fourier transforms of f and g .
- b) Determine whether or not a function $h \in L^1(\mathbb{R})$ exists such that

$$\int_{-\infty}^{\infty} h(t-s)f(s) ds = g(t)$$

for all $t \in \mathbb{R}$, and, if yes, provide an example of such an h .

Please, turn over!

5. a) Prove that the series

$$\sum_{n=1}^{\infty} \frac{x^2}{n(1+nx^4)}$$

converges uniformly on $\mathbb{R}$.

- b) Compute

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \int_0^4 \frac{x^2 \cos(kx)}{n(1+nx^4)} dx.$$

6. Decide whether or not a function with the respective properties exists. If yes, provide an example; if not, prove that no such function can exist.

- a) $f \in C(\mathbb{T})$ with $\hat{f}(n) = \frac{1}{1+n^2}$ for all $n \in \mathbb{Z}$.

- b) $f \in L^1(\mathbb{R})$ with

$$\hat{f}(\omega) = \begin{cases} \frac{1}{1+\omega^2}, & \omega > 0, \\ 0, & \omega \leq 0. \end{cases}$$

- c) $f \in L^1(\mathbb{R})$ differentiable, bounded with $f \neq 0$ and $f * f = f$.